

Knowing when not to automate

Reframing an automation request in surgical-neuroimaging QC so the human reviewer stays the decision-maker

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Context

The product turns a patient's MRI into a map of white-matter tracts and functional brain networks, for a neurosurgeon planning a resection around a tumour. It is regulated software. Before any map reaches a surgeon it is quality-reviewed — and the reviewers doing the first pass are operations staff **without a neuro background**; a clinical radiologist gives the final read. Most incoming cases are hard: deep-seated, infiltrative tumours. Review is the last place an error is caught before an operating room.

The ask

Reduce reviewer burden: have the software **automatically find the tumour** and **automatically select the tracts and networks "relevant" to it**, so reviewers only look at those.

Why the obvious build fails

I tested the automation on a representative hard case *before* committing engineering to it. A standard intensity method for locating the tumour flagged roughly a tenth of the whole brain as candidate lesion, with no left-right signal — on a case whose tumour was unambiguously left-sided.

~9% of brain volume flagged · 0 lateralising signal

Not a tuning gap. The most common cases are exactly the infiltrative type whose margins blend into normal tissue — so automated detection is least reliable precisely where it is needed most. Automating this step moves the least-reliable judgment onto the least-qualified reviewer.

The deeper problem — "relevant" is a clinical judgment, not a geometric one

Even with a perfect tumour outline, ranking tracts and networks by proximity to the tumour does the wrong thing twice. It surfaces first the activations sitting **inside** the tumour — which is exactly where a well-documented false-signal artifact lives — anchoring a non-specialist onto the artifact. And it demotes tracts **pushed away** from their normal path by the tumour's mass — often the ones a surgeon most needs to avoid. Proximity correlates with the wrong things.

The reframe: the tool computes geometry; the human makes every call.

The software never detects the tumour, never labels a tract "broken," never decides relevance. Tumour location becomes an **input**, supplied once by someone clinically competent. The tool then does deterministic bookkeeping — and presents **every** output, ranked, never a filtered subset, because dropping the "irrelevant" ones would be making the exact clinical call I removed.

STAGE A · NEEDS NO TUMOUR LOCATION

Did every expected output generate? Any empty or truncated? Which have low pipeline confidence? Left-right anomalies? Pure completeness & integrity — zero clinical judgment.

STAGE B · WHEN A LOCATION IS SUPPLIED

Proximity ranking to help the reviewer prioritise. Every distance is stamped a **cue to check, not a measurement** — the outputs sit on a known, unresolved registration error, and the non-specialist can't sense that error bar.

What I built

A working prototype that ingests a result folder, runs the checks, and **pre-fills the reviewer's existing QC sheet** with structural cues and a suggested review order — leaving every clinical yes/no blank for the human — and writes a **per-case audit trail** that becomes the documented record of what happened to each case. The internal record plus the radiologist's clinical read form one traceable file per case.

What I deliberately left open — the honest version names what isn't solved

- 01

Workflow ordering. The tumour location needs clinical competence, which conflicts with a fully non-specialist first pass — so either the radiologist's note moves earlier, or Stage B only appears after the radiologist has touched the case. A real trade-off, not a detail.
- 02

Regulatory classification. Because the tool influences what gets flagged and regenerated, whether it sits inside the medical-device boundary is a genuine design-control question.
- 03

Automation anchoring. A "review first" cue could pull a non-specialist toward the tumour region and away from the displaced tract — a training-and-audit risk software can't fix.
- 04

No validated impact yet — so the piece claims none.

HOW I'D PROVE IT WORKS

Run the tool silently on ~20 already-reviewed cases; measure how often its top-ranked outputs match the ones the radiologist actually marked clinically relevant. Poor overlap falsifies the proximity proxy — learned retrospectively, on old cases, not on a surgery morning.

What this demonstrates

Evidence over assumption Killed the requested build with data before it consumed engineering.	Regulatory-aware scoping Design-control boundary treated as a first-order product question.	Human-in-the-loop by design The reviewer stays the decision-maker as architecture, not a disclaimer.	Honest about the unproven Ships the open questions and a falsifiable validation plan with the solution.
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